

# dsupdt

Periodic Dataset Updates for the GDEX Servers

*Download → Build → Archive, scheduled and automated*

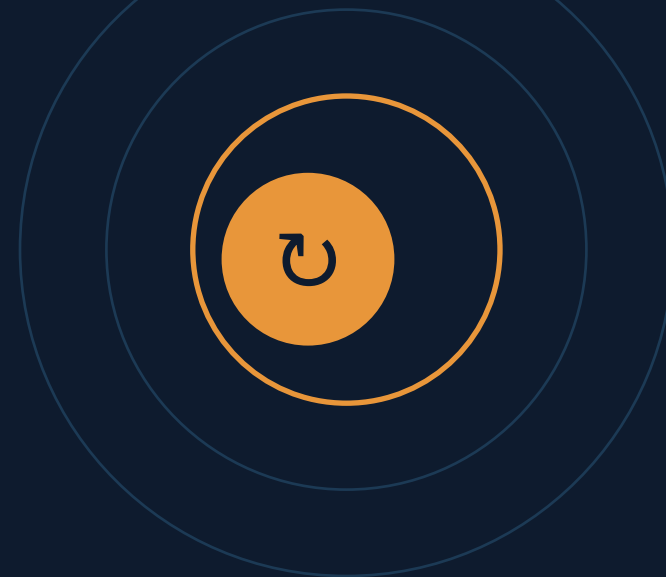
Developer Configuration

General Usage

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Companion utility to `dsarch` • orchestrated by the `dscheck` daemon



# What is dsupdt?

**dsupdt** performs **periodic dataset updates** — it retrieves data from remote servers, assembles archive-ready local files, and hands them to **dsarch** for archiving onto the GDEX Servers.

## Configured per dataset in GDEXDB

All operational datasets run under **dsupdt / dsarch**; configuration spans three record types.

Edit records from the CLI (

**-GA / -SA**) or the web Config Editor (next slide).

Only the owning specialist may run an update record (override with

**-MD**).

Input files must be named

**dNNNNNN . \*** to guard against the wrong dataset.



### Configure

update controls, local & remote file records



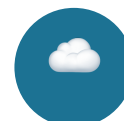
### Download

copy / fetch server files to a working area



### Validate & Build

run specialist routines, tar & compress



### Archive

hand local files to dsarch → GDEX



### Clean

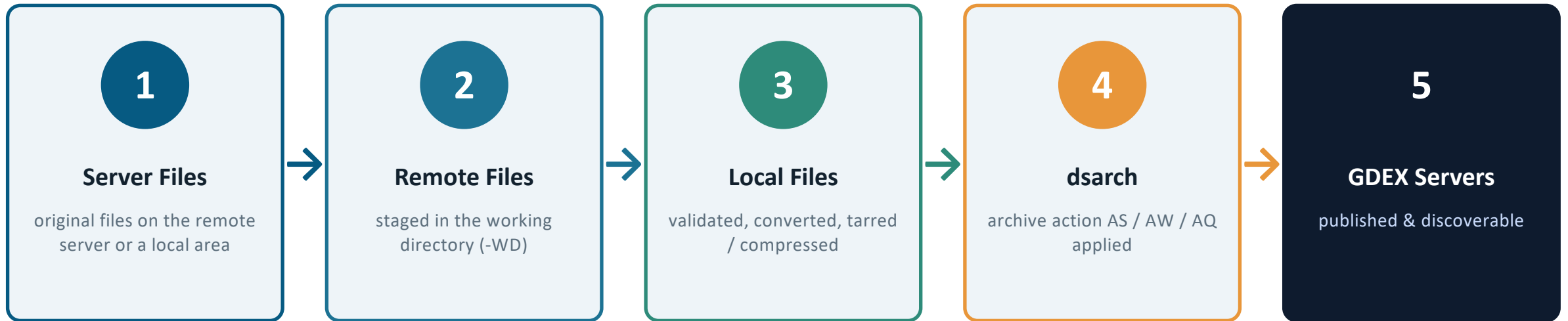
remove temporary files after success



### Check

test remote-file availability

# The Five-Stage Update Pipeline



**One command runs the whole chain.** `-UF` (`-UpdateFile`) performs all stages end-to-end. The individual steps `-DR`, `-BL`, `-PB`, `-AF`, `-CF` let you drive each stage separately.

# Three Record Types in GDEXDB

## Update Control

dcupdt    -SC / -GC

- Schedules when updates run
- Frequency, offset, retry, valid interval
- Email & error controls
- Launched by the dscheck daemon

## Local File

dlupdt    -SL / -GL

- Minimum config for an update
- Local file name & dsarch action
- Download / build / clean commands
- Tracks next end date/hour

## Remote File

drupdt    -SR / -GR

- Only when remote ≠ local name
- Many remotes → one local file
- Server file & download order
- Time interval for sub-periods

**Get → edit → Set:** -GA (-GetAll) dumps all three sections; -SA (-SetAll) applies an edited file back to GDEXDB.

# Anatomy of a Command

```
dsupdt [-DS dNNNNNN] <Action> [Mode...] [Info...]
```

## Action

One per run

Selects the task. Takes no value. E.g. -UF, -GC, -SL, -AF. Some actions imply others.

## Mode

Zero or more

Modifies behavior; takes no value. E.g. -MU, -FU, -RA, -NE, -GZ.

## Info

Carries data

Supplies values: dataset, indices, dates, file names, commands. E.g. -CI, -ED, -LF.

**Notation & filters** (A|B) short|long form • option names are case-insensitive, values are not • Get-style filters accept ! < > <> • the first -DS may omit its option name.

# Quick Start

## Show a single option's help

```
dsupdt -h -UF
```

## Get update control records

```
dsupdt d540000 -GC
```

## Get local file records

```
dsupdt d540000 -GL
```

## Run download–build–archive

```
dsupdt d540000 -UF
```

## Process all elapsed periods

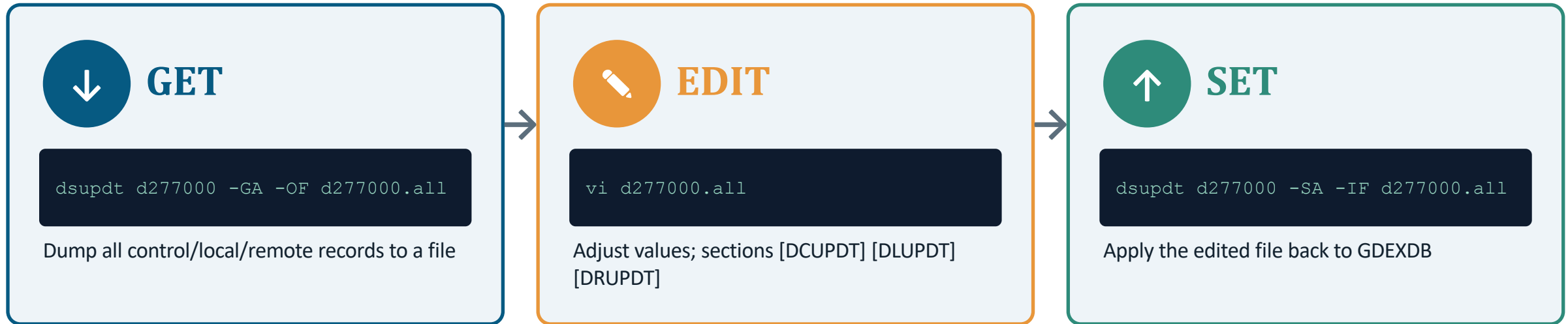
```
dsupdt d540000 -UF -MU -IE
```

## Check remote-file availability

```
dsupdt d540000 -CU
```

**Tip:** run **dsupdt** with no arguments to page the full guide through **more**. Omit **-DS** on Get actions to list records across all of your datasets.

# Configuration Workflow



## Input file (-IF) format

|                   |                                              |
|-------------------|----------------------------------------------|
| # ...             | comment line                                 |
| Dataset<=>d277000 | single-value assignment ( -ES sets '<=>' )   |
| SC<!>             | Action / Mode marker ( -AO sets '<!>' )      |
| col1<:>col2<:>    | tabular header row ( -DV sets '<:>' )        |
| dNNNNNNN.*        | file name must start with the dataset number |

# Web Config Editor

The same three record types can be edited in a browser — no **-GA** / **-SA** Get/Edit/Set round-trip. Open the **RDAMS PostgreSQL Configuration Editor** and expand the **DSUPDT** menu.

```
gdex.ucar.edu/rda_pg_config > DSUPDT > Update Control | Local File | Remote File
```

## Update Control

```
dsupdt_control.php
```

**dcupdt** • **-SC** / **-GC**

frequency, next-due, actions, e-mail & error controls

## Local File

```
dsupdt_localfile.php
```

**dlupdt** • **-SL** / **-GL**

file names, patterns, build / convert commands, archive action

## Remote File

```
dsupdt_remotefile.php
```

**drupdt** • **-SR** / **-GR**

remote & server names, download command & order



**Split-pane editor:** pick a dataset & record on top, edit fields below. A record is **locked** while open; add, edit, save, or delete, then release. Writes go straight to GDEXDB — identical to the CLI.



# Update Control Record (-SC)

Schedules automatic actions launched by **dscheck**. New record: index 0 + **-NC**. Existing index is modified in place.

| Opt | Long name     | Purpose                                 |
|-----|---------------|-----------------------------------------|
| -AN | ActionName    | dsupdt task: UF, DR, BL, PB, CL, CU     |
| -FQ | Frequency     | how often, e.g. 1W, 1M, 6H, 1M/3        |
| -CO | ControlOffset | delay after period end, e.g. 2D10H      |
| -CT | ControlTime   | next scheduled run YYYY-MM-DD HH:NN:SS  |
| -RI | RetryInterval | wait before retry after a failure       |
| -VI | ValidInterval | how long remote files stay valid        |
| -PI | ParentIndex   | run only after parent control completes |
| -UC | UpdateControl | preset modes: B C E F G M N O Y Z       |
| -MC | EMailControl  | A / S / E / B / N report level          |
| -EC | ErrorControl  | I ignore / Q quit / N normal            |
| -KF | KeepFile      | S / R / B / N which files to keep       |
| -HN | HostName      | restrict/allow processing hosts         |

# Local File Record (-SL)

The **minimum** configuration to update a file. Link to a control via **-CI** so its scheduled action runs automatically. New: **index 0 + -NL**.

| Opt | Long name       | Purpose                           |
|-----|-----------------|-----------------------------------|
| -LF | LocalFile       | name, may embed temporal patterns |
| -AN | ActionName      | dsarch action: AS, AW, AQ         |
| -OP | Options         | options string passed to dsarch   |
| -DC | DownloadCommand | fetch/copy/generate the file      |
| -FQ | Frequency       | data cadence, e.g. 1M, 1W, 6H     |
| -ED | EndDate         | data end date of next update      |
| -EH | EndHour         | end hour (sub-daily cadence)      |

| Opt | Long name     | Purpose                      |
|-----|---------------|------------------------------|
| -DI | DuelInterval  | end → ready lead time        |
| -VI | ValidInterval | re-check / hold-open window  |
| -AT | AgeTime       | min remote-file age to fetch |
| -WD | WorkDir       | staging dir (root \$UPDTWKP) |
| -PR | ProcessRemote | user validate/convert cmd    |
| -BC | BuildCommand  | user build cmd (vs tar)      |
| -CL | CleanCommand  | remove temp files after      |

**Patterns** in -LF / -RF / -DC / -DE (default delimiters < >) are substituted at update time, e.g. **file.<YYYYMM>.ext** → **file.200710.ext**.

# Remote File Record (-SR)

Add a remote record only in these three cases:



**Different name**

remote file name differs from the local file name



**Many → one**

several remote files are tarred into one local file



**Multiple servers**

one file available from several servers (-DO order)

| Opt | Long name       | Purpose                                              |
|-----|-----------------|------------------------------------------------------|
| -LI | LocalIndex      | local record this remote belongs to (required)       |
| -RF | RemoteFile      | remote name(s); join many with ':' ; '!' = shell cmd |
| -SF | ServerFile      | name on server if different from remote name         |
| -DC | DownloadCommand | overrides the local record's command                 |
| -DO | DownloadOrder   | try lowest index first, then ascending               |
| -TI | TimeInterval    | one name per sub-period, e.g. 6H, 5D                 |
| -BT | BeginTime       | window start (with -TI)                              |
| -ET | EndTime         | window end (with -TI)                                |

# The Update Action Family

**-UF**

## Update File

end-to-end: download → build → archive → clean

**-DR**

## Download Remote

fetch/copy/generate remote files

**-BL**

## Build Local

validate, convert, tar/compress

**-PB**

## Process Both

download + build in one step

**-AF**

## Archive File

hand local files to dsarch

**-CF**

## Clean File

remove temporary working files

**-CU**

## Check Update

test remote-file availability

**-UL**

## Unlock

clear locks from aborted runs

**-UF bundles the four steps.** Stepping manually is useful for debugging or re-running a single stage. All actions accept **-CI** / **-LI** / **-LF** / **-XO** to target records, and **-ED** / **-EH** / **-CD** / **-CH** to override dates at run time.

# -UF Deep Dive: One Command, Full Cycle

## For each due local file record

1. Download / copy the server file
2. Validate & convert (-PR) into a remote file
3. Build the local file (tar / compress or -BC)
4. Archive via dsarch using the record's action
5. Clean temporary files (-CL)
6. Advance data end date/hour & next-due times

| Mode/Info | Effect                                          |
|-----------|-------------------------------------------------|
| -MU       | process every elapsed period (else newest only) |
| -FU       | force at least one update even if not due       |
| -CP       | allow the current, not-yet-complete period      |
| -RA       | re-archive (pass -RA to dsarch)                 |
| -RD       | re-download even if the file exists locally     |
| -MO       | only periods not yet archived                   |
| -IE       | skip failures & continue (with -MU)             |
| -QE       | stop the dataset on first error                 |
| -PL       | fork N child processes for parallel records     |
| -CC       | add carbon-copy email recipients                |

```
dsupdt d337000 -UF -MU -IE -CC schuster
```

# Archiving & Content Metadata

Once a local file is built, **dsupdt** hands it to **dsarch** to archive onto GDEX, then optionally gathers content metadata so the new data is discoverable.



## Archive to GDEX — dsarch

dsupdt calls dsarch with the record's action (AS / AW / AQ). -OP appends extra dsarch options; -RA forces re-archiving. This is the step that publishes each file.



## Gather content metadata — gatherxml (-GX)

Optional. When -GX is set in the local record's Options, dsupdt runs gatherxml on the just-archived file to index its variables and spatial / temporal coverage.



## Refresh the search index — scm

After archiving, dsupdt runs the search-metadata command (scm -d DS -r w) to summarize / refresh the dataset's web tables so the new content is searchable.

**Metadata steps are non-fatal:** a gatherxml or scm failure is reported in the e-mail but never blocks or retries the archive.

# Temporal & Generic Patterns

Placeholders in file names, download commands, and descriptions are substituted at update time. Default delimiters < > (change with -PD). By default patterns use the period's **end** date/hour.

| Pattern        | Basis    | Meaning                                        |
|----------------|----------|------------------------------------------------|
| <YYYYMM>       | end date | 4-digit year + 2-digit month → 200710          |
| <YYYY/MM/DD>   | end date | path-style date components                     |
| <CYYYY.MM.DDC> | current  | wrap in C ... C to use the current date/hour   |
| <BYYYY.MM.DDB> | begin    | wrap in B ... B to use the period's begin time |
| <M*M>          | fraction | per-fraction id: C=A,B,C c=a,b,c N=1,2,3       |
| <HH>           | end hour | 2-digit hour for sub-daily cadence             |
| <P0> <P1>      | runtime  | generic values supplied via -GP                |

## Example

```
-LF uv.<YYYYMM>.bln
```

```
end date 2007-10-31
```

↓ **becomes**

```
uv.200710.bln
```

**Dynamic work dir:** when a server path carries a date the remote name lacks, embed that date in -WD so different dates don't overwrite each other.

# Custom Commands with '!'

Some fields are shell commands by design (**-DC** / **-BC** / **-CL**). Any value field can also become a command by prefixing it with **'!'** — dsupdt runs it and uses the stdout as the value. Temporal patterns are substituted first.

**-DC****Download Command**

How each remote file is fetched from the server (wget, cp, mftp, ...).

**-BC****Build Command**

External utility that assembles / converts the local file after download.

**-CL****Clean Command**

Post-archive cleanup command for temporary or intermediate files.

**-RF '!...'****Remote File**

Name resolved at run time — e.g. pick the newest matching server file.

**-LF '!...'****Local File**

Compute the local file name from a script instead of a fixed pattern.

**-DE '!...'****Description / note**

Generate a data note dynamically from command output.

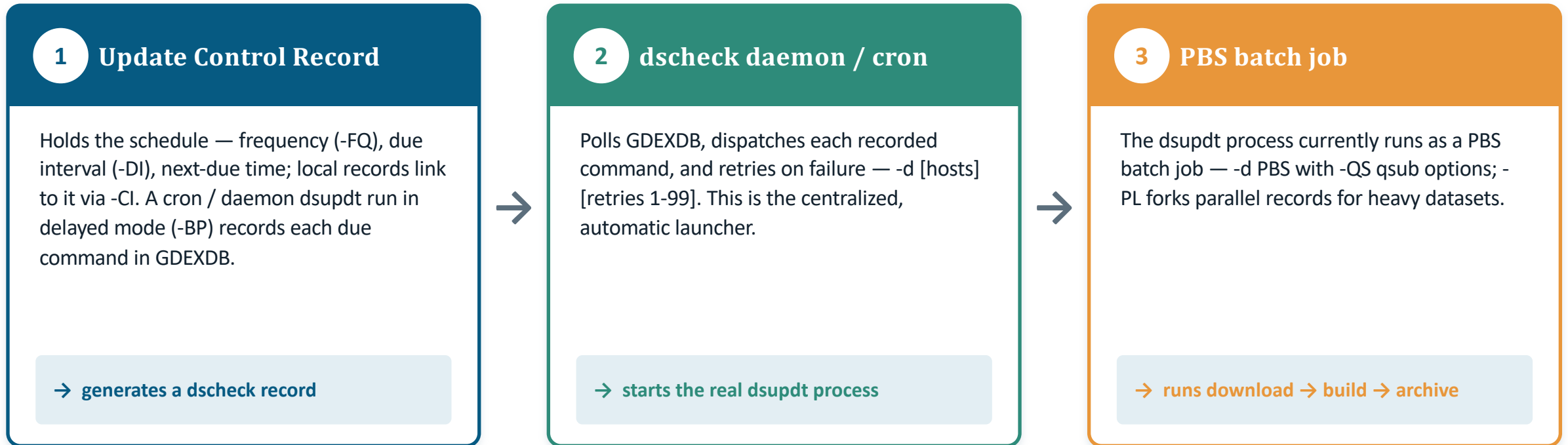
**Example — pick the newest matching file at update time**

```
dsupdt d540000 SR -LI 1 -RF '!ls -t /glade/incoming/<YYYYMM>*.grib | head -1'
```



# Scheduling & Execution

In production the whole chain is automated — currently under **PBS batch control**:



**Direct / manual:** run `dsupdt -UF` yourself — or from a plain cron job on a specific host — for one-off runs, re-archiving, and debugging. No control record or dscheck needed.

# Valid Interval & Multi-Period Updates

## **-FQ** Frequency

the size of one update period (1D, 6H, 1M, ...)

## **-MU** MultipleUpdate

process every elapsed period, not just the newest

## **-VI** ValidInterval

look back this far to re-check / re-archive late data

## **-DI** DueInterval

lead time from period end until data is expected

**-IE** / **-QE** / **-EC** govern how errors interact with multi-period runs: skip & continue, quit, or normal.

## Re-check window (VI = 4D, FQ = 1D)



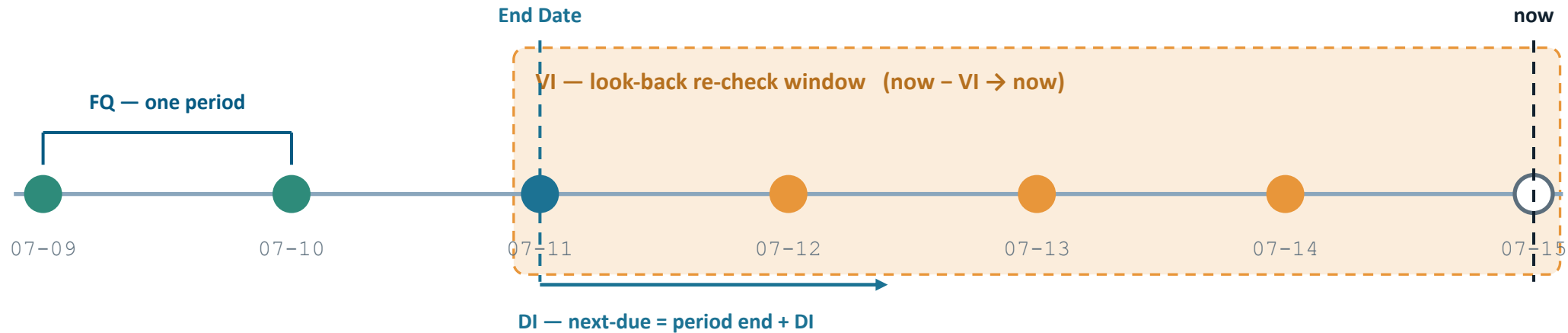
Each period is re-checked. Already-archived periods report “no newer file”; a newer source file triggers **re-archive**. VI = 0 → single period, no newer-file check.

## Partial updates

With **-MR Y**, a many-to-one build tolerates missing remotes. If **-VI** is also set, the update is held open until the files arrive or the window expires.

# How the Temporal Intervals Fit Together

Data advances one **Frequency (FQ)** per period. A single **-MU** run walks every elapsed period inside the **Valid Interval (VI)** look-back window, while the **Due Interval (DI)** sets when each period becomes due.



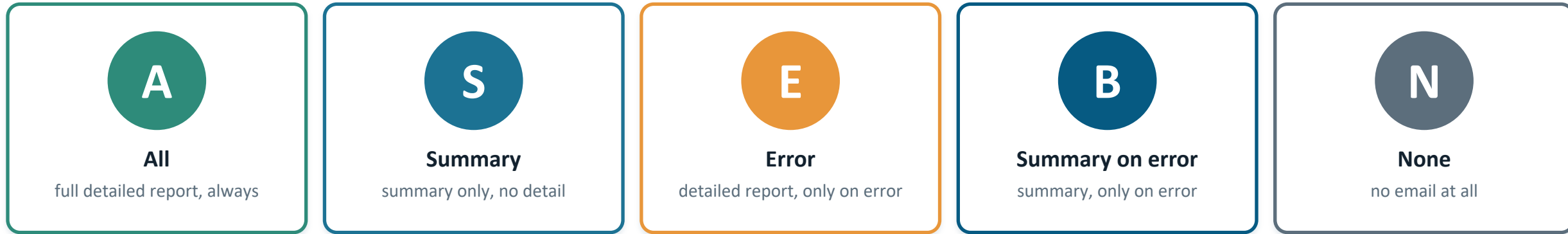
● Archived earlier
 ● Re-checked — no newer source
 ● Archived / re-archived this run
 ○ Not yet due (waiting DI)

## ControlTime — when the job itself runs

**-CT** sets the scheduled run time (YYYY-MM-DD HH:NN:SS). After a success it advances by control **-FQ** + **-CO** offset; a failure retries after **-RI**. e.g. `dsupdt d609000 -SU -CT "2026-07-11 09:00:00" -FQ 1D -CO 9H`

**Each run re-checks every period in the VI window:** already-archived periods report “no newer file”, newly available periods are archived, and periods still within DI of now are left for the next run. *VI = 0 → only the most recent period.*

# Email Reporting (-MC / -EMailControl)



## Report sections

|         |                                                                                                    |
|---------|----------------------------------------------------------------------------------------------------|
| Error   | real errors — survives in every mode; raw failures are prefixed with the failing command / dataset |
| Summary | headline count + rolled-up “N files ARCHIVED” range lines                                          |
| Detail  | per-file lines; consecutive no-op re-checks collapse into one range line                           |

**Related:** `-EE` (=E), `-SE` (=S), `-NE` (=N), `-CC` adds carbon-copy recipients.

# Key Mode Options at a Glance

|                                              |                                               |                                               |
|----------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| <b>-MU</b> process all elapsed periods       | <b>-FU</b> force one update even if not due   | <b>-CP</b> allow current, not-yet-due period  |
| <b>-MO</b> only not-yet-archived periods     | <b>-RA</b> force re-archive via dsarch        | <b>-RD</b> re-download existing remote file   |
| <b>-CN</b> re-download if changed on server  | <b>-RE</b> reset end date/hour from file time | <b>-IE</b> skip errors, continue (with -MU)   |
| <b>-QE</b> quit dataset on first error       | <b>-GZ</b> use GMT as controlling time        | <b>-NY</b> skip Feb 29 in leap years          |
| <b>-KR</b> keep remote file (copy, not move) | <b>-KS</b> keep server file (copy, not move)  | <b>-UB</b> use period begin time for patterns |
| <b>-UT</b> force end/next-due time advance   | <b>-BG</b> background; suppress screen output | <b>-NE</b> suppress notification email        |

*Most modes can be preset in the control record via -UC (single letters) and -KF / -MC / -EC.*

# Environment & Prerequisites



## Under dsarch/dsupdt

All operational datasets run under dsupdt/dsarch control; dsupdt hands its local files to dsarch for archiving onto GDEX.



## \$UPDTWKP

Root for working dirs. If set to /lustre/gdex/work, then \$UPDTWKP/zji/icoads resolves under it; if unset, -WD is used as given.



## Ownership

Only the owning specialist may run a record. Use -MD to override, or -LN to run on behalf of another (auto-adds -MD to dsarch).



## dNNNNNN.\*

Input files must start with the dataset number and match -DS — a guard against operating on the wrong dataset.



## -PL parallelism

Fork N child processes, each handling one record, for datasets with many independent, time-consuming updates.



## -DB / logs

Debug logging via -DB level [path] [file]; default log path \$DSSHOME/dssdb/log, file pgdss.dbg.

# Troubleshooting & Recovery

## Stuck lock

A crashed run may leave PID/host on a record, blocking reprocessing on another host.

```
dsupdt -UL -CI 57  
dsupdt -UL -LI 33
```

## Is data ready?

Test remote-file availability before committing to an update.

```
dsupdt d540000 -CU -CA
```

## Re-run a period

Substitute a different date to re-archive or recover a deleted action.

```
dsupdt d540000 -UF -CD 2026-07-01 -RA
```

## Retry on failure

Control records retry automatically after -RI; -EC sets error behavior.

```
dsupdt d540000 -SC -CI 5 -RI 12H -EC N
```

## Recap

# From Config to Archive

- 1 **Configure** three record types: control (dcupdt), local (dlupdt), remote (drupdt)
- 2 **Run** -UF drives download → build → archive → clean in one command
- 3 **Automate** link to a control record; dscheck dispatches, retries, and reports
- 4 **Tune** patterns, valid interval, parallelism, and email controls per record

**Learn more** `dsupdt -h <OPT>` • full guide: [dsupdt.usg](https://dsupdt.usg) • docs: [gdex-docs-dsupdt.readthedocs.io](https://gdex-docs-dsupdt.readthedocs.io) • repo: [github.com/NCAR/rda-python-dsupdt](https://github.com/NCAR/rda-python-dsupdt)





# Any Questions?

*Thank you*

`dsupdt -h <OPT>` • [dsupdt.usg](https://dsupdt.usg) • [github.com/NCAR/rda-python-dsupdt](https://github.com/NCAR/rda-python-dsupdt)